

Advisor Portal -- Technical Documentation

System Architecture

Overview

The Advisor Portal is a real-time workspace for service center advisors to manage the complete vehicle service lifecycle from lead acceptance to car delivery. It operates within the Performance Intelligence System (PIS) and integrates with all workshop modules.

Tech Stack

- **Frontend:** Next.js 16, React 19, TypeScript
- **Backend:** Next.js API Routes, PostgreSQL 16
- **Real-time:** Auto-refresh polling (5s with pending offers, 30s idle)
- **Auth:** Session-based JWT, PIS role-based access

File Structure

```
apps/web/app/company/[companyId]/pis/advisor-portal/
  page.tsx                                -- Main advisor portal page (client component)

apps/web/app/api/company/[companyId]/pis/
  advisor-portal/route.ts                 -- Main data API (leads, calls, workflow, KPIs)
  lead-distribution/
    route.ts                             -- Lead routing engine
    accept/route.ts                      -- Lead acceptance endpoint
    config/route.ts                      -- Distribution config

packages/ai-core/src/pis/
  leadDistribution/
    engine.ts                            -- PIS routing, cascade, accept logic
    autoAssign.ts                        -- Auto-assign trigger on car check-in
    repository.ts                        -- Queue CRUD operations
  advisorRepository.ts                   -- Advisor scoring (7 weighted metrics)
  configRepository.ts                   -- PIS config + service charges
  service.ts                            -- Master service aggregator
```

Data Flow

1. Lead Assignment (Auto via PIS)

```
Car Check-In (lead_status = car_in)
|
v
triggerAutoAssignOnCarIn() [autoAssign.ts]
```

```

|
v
Guard: already assigned? already in queue? -> SKIP
|
v
routeLead() [engine.ts]
|
v
Fetch advisors sorted by composite_score DESC
|
v
Filter out locked advisors (handling another lead)
|
v
Offer to best available advisor
|
v
Create pis_lead_queue entry (status: offered)
|
v
Set locked_until = now + accept_window (tier-based: 5/7/10 min)

```

2. Lead Acceptance

```

POST /api/company/{id}/pis/lead-distribution/accept
|
v
acceptLead() [engine.ts]
|
v
Update queue: status=locked, accepted_at=now, locked_until=now+120min
|
v
applyAdvisorToLead() [autoAssign.ts]
|
v
Set leads.agent_employee_id, assigned_user_id, assigned_at
|
v
Log lead_event: advisor_accepted

```

3. Cascade on Timeout

```

checkExpiredOffers() [engine.ts]
|
v
For each expired offer:
  Apply 5-point penalty to timed-out advisor
  Log history: timeout

```

```

|
v
cascade_count < max_cascade_attempts (5)?
YES -> Offer to next best advisor
NO -> Escalate to manual queue

```

API Endpoints

GET /api/company/{companyId}/pis/advisor-portal

Query: ?advisorUserId={userId}

Returns:

```

{
  leads: Array<{
    id, customer_name, customer_phone, lead_status, lead_type,
    checkinAt, carPlateNumber, carModel, branchName, serviceType,
    customerId, customerWalletAmount,
    workflow: {
      inspection: { id, status } | null,
      estimate: { id, status, grandTotal } | null,
      jobCard: { id, status, startAt, completeAt, finalInspectionAt, carWash, ...
    } | null,
    parts: { orderedCount, receivedCount, items: [...] } | null,
    invoice: { id, status, grandTotal, invoiceNumber } | null,
    qualityCheck: { id, status } | null,
    gatepass: { id, status } | null,
    preInspectionForm: { status, submittedAt } | null,
  }
}>,
  calls: Array<{ id, direction, from_number, to_number, status, duration_seconds
}>,
  pendingOffers: Array<{ queueId, leadId, tier, offeredAt, lockedUntil,
pipelineValue, customerName, ... }>,
  kpis: {
    totalLeads, convertedLeads, carInLeads, totalCalls, answerRate,
    revenue, invoiceCount, carOutCount, todayCollection
  }
}

```

POST /api/company/{companyId}/pis/lead-distribution/accept

Body: { queueId: string }

GET /api/company/{companyId}/workshop/service-charges

Query: ?leadId={leadId} **Returns:** { inspectionFee, recoveryPickupFee, recoveryDropoffFee, labourCharge, currency }

POST /api/company/{companyId}/workshop/invoices

Body: { estimateId, serviceCharges: { inspectionFee, recoveryPickupFee, recoveryDropoffFee, labourCharge }, autoSettleOnConvert, autoCarOutOnAutoPaid }

POST /api/customers/{id}/wallet/transactions

Body: { companyId, amount, paymentMethod, paymentDate, paymentProofFileId }

POST /api/company/{companyId}/workshop/invoices/{invoiceId}/pay

Body: { paymentMethod: "wallet" }

POST /api/company/{companyId}/workshop/gatepass

Body: { invoiceId, handoverType: "branch" | "dropoff_recovery" }

PATCH /api/company/{companyId}/workshop/gatepass/{gatepassId}

Body: { status: "released", customerSigned: true, supervisorApprovedAt, finalNote }

Database Tables

Core Tables

Table	Purpose
leads	Master lead record with agent assignment
pis_lead_queue	Lead offer/acceptance queue
pis_lead_queue_history	Queue action audit trail
pis_advisor_scores	Advisor performance scores
pis_config	PIS configuration (weights, thresholds, service charges)

Workflow Tables

Table	Purpose
inspections	Vehicle inspection records
estimates	Cost estimates with items
job_cards	Work execution tracking
part_quotes	Vendor quotes for parts
purchase_orders	Parts procurement
invoices	Customer billing
gatepasses	Car delivery/handover

Table	Purpose
quality_checks	QC records (optional, final inspection used)

Supporting Tables

Table	Purpose
line_items	Inspection findings / parts needed
estimate_items	Estimate line items with pricing
invoice_items	Invoice line items
purchase_order_items	PO line items
inventory_movements	Stock movements (GRN)
inventory_stock	Current stock levels
lead_bookings	Appointment bookings
pre_inspection_form_requests	Customer pre-inspection forms
customer_wallet_transactions	Wallet topup records

Scoring Algorithm

Composite Score (0-100)

Metric	Weight	Description
Conversion Rate	25%	Leads converted / total leads
Gross Profit %	20%	(Revenue - Cost) / Revenue
Revenue	15%	Normalized total revenue
SLA Compliance	15%	Estimate approval rate
Customer Satisfaction	10%	Call completion / health scores
Call Quality	10%	Completed calls %
FOC Penalty	-5%	Free-of-charge rate (negative)

Tier Assignment

Tier	Score	Accept Window
1 (ELITE)	>= 80	10 minutes
2 (STANDARD)	>= 50	7 minutes
3 (LOW)	< 50	5 minutes

Configuration (PIS Admin)

service_charges

```
{
  "inspection_fee": 150,
  "recovery_pickup_fee": 200,
  "recovery_dropoff_fee": 200,
  "labour_charge": 0,
  "currency": "AED"
}
```

lead_distribution

```
{
  "tier1_accept_window_min": 10,
  "tier2_accept_window_min": 7,
  "tier3_accept_window_min": 5,
  "lock_duration_min": 120,
  "no_call_penalty_points": 5,
  "max_cascade_attempts": 5
}
```

Error Handling

Scenario	Handling
No advisors available	Queue stays pending, retries on next check
Accept window expired	Auto-cascade with 5-point penalty
Max cascades exceeded	Escalate to manual queue
Wallet insufficient	Show shortfall, allow topup
Invoice creation fails	Show error, keep estimate
Gatepass release fails	Show error, retry available

Security

- Session-based authentication required
 - Company-scoped data isolation
 - Role-based permission checks on mutations
 - Auto-refresh uses authenticated API calls
 - File uploads via `/api/files/upload` with auth
-

Version: 1.0 | Date: 2026-03-31 | Global ERP System